A Controlled Reevaluation of Coreference Resolution Models

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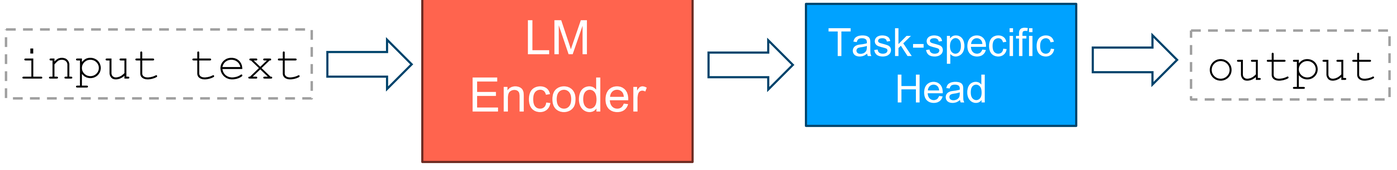
McGill University Mila – Quebec AI Institute *equal contribution

Q: ***What accounts for increases in the accuracy of coreference resolution models?***

A: *Controlling for the choice of language model (LM) reduces most of the increase in F1 score reported in the past five years.*

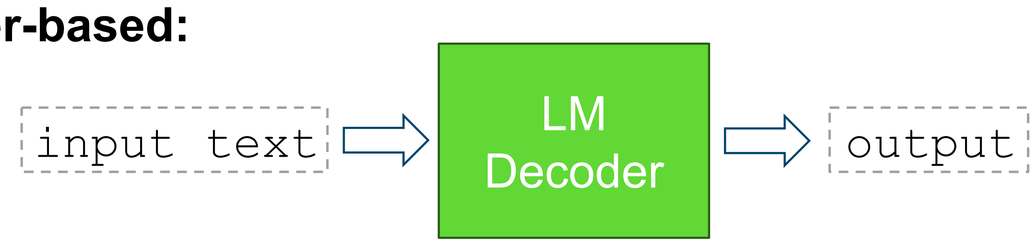
Coreference Resolution Models

Encoder-based:



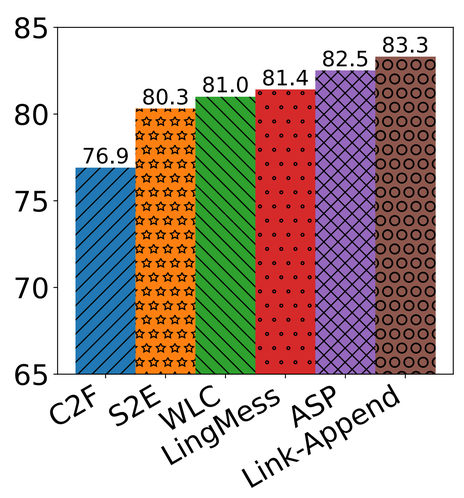
e.g.,
- **C2F (**Coarse-to-Fine; Lee et al., 2018)
- **S2E** (Start-to-End; Kirstain et al., 2021)
- **WLC** (Word-Level; Dobrovolskii, 2021)
- **LingMess** (Otmazgin et al., 2023)

Decoder-based:



e.g.,
- **ASP (**Liu et al., 2022)
- **Link-Append** (Bohnet et al., 2023)

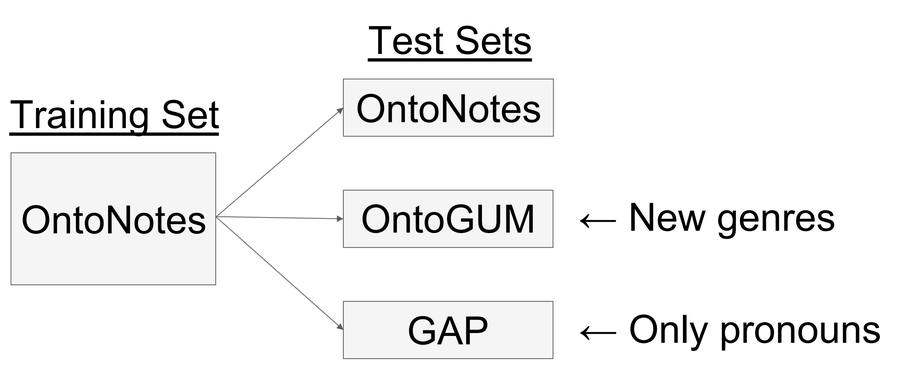
Context: Reported accuracy is steadily increasing with each new approach.

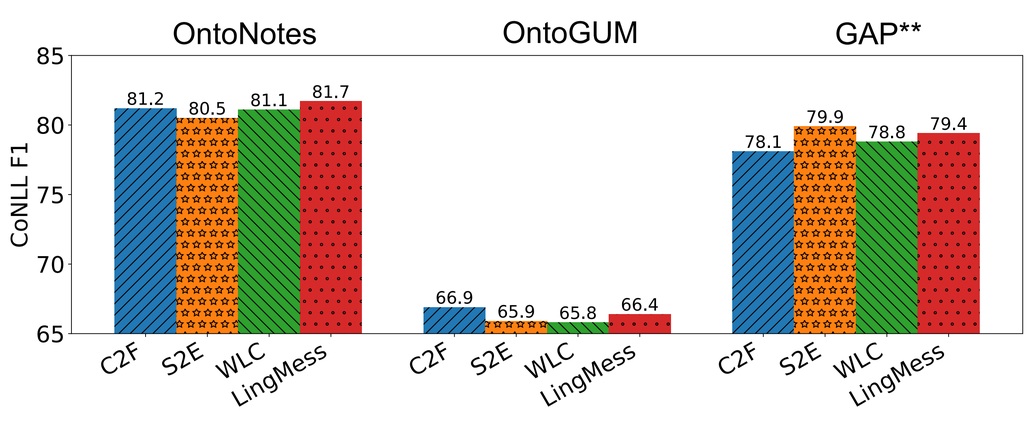


CoNLL F1 scores on OntoNotes test.

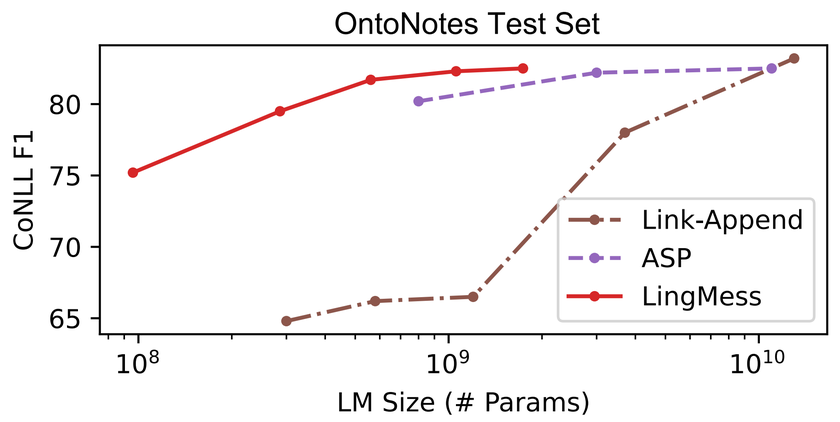
Models are ordered by the earliest upload date of the respective paper.

Method: We reimplement and compare coreference resolution models, controlling for the choice of LM and hyperparameter decisions.





Results



When we control for the potential confounds of **size(** LM Encoder **)**, **size(** Task-specific Head **)**, **number of training epochs**, and **input length**, we find that the gap between encoder-based models shrinks. The oldest model, C2F, is most accurate on the OntoGUM dataset.

**Instance-level F1.

When size(LM Encoder) ≈ size(LM Decoder), **encoder-based models outperform decoder-based models**; *e.g.,* at 1.5B params, LingMess achieves the same F1 score as ASP with 11B params.

Example instance: Highlighted text marks coreferring mentions. Underlined is a precision error of a 13B param Link-Append model.

> Former President Clinton drew cheers from admirers at a US Air Force base outside Washington today before departing for his home in New York, and although Mr. Clinton is out of office, he says he'll still be around. […] Prior to leaving office today Mr. Clinton issued more than one hundred presidential pardons. […]

Conclusion:
- *We reimplement five recent coreference resolution models.*
- *We find that the choice of LM may be responsible for recent improvements in coreference resolution accuracy.*

Additional details in the paper:

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